

DHCP Snooping

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Server valide

Server DHCP valide

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: DHCP_valide

Default Gateway: 192.168.1.254

DNS Server: 0.0.0.0

Start IP Address: 192 168 1 100

Subnet Mask: 255 255 255 0

Maximum Number of Users: 10

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	192.168....	255.255....	0	0.0.0.0	0.0.0.0
DHCP_valide	192.168....	0.0.0.0	192.168....	255.255....	10	0.0.0.0	0.0.0.0

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PC 1

IP Configuration	
☒ DHCP	☐ Static DHCP request successful.
IPv4 Address	192.168.1.101
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.254
DNS Server	0.0.0.0

PC2

IP Configuration	
☒ DHCP	☐ Static DHCP request successful.
IPv4 Address	192.168.1.102
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.254
DNS Server	0.0.0.0

IPv6 Configuration

Server invalide

Server DHCP de la personne malveillante

Physical Config **Services** Desktop Programming Attributes

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DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: DHCP_invalide

Default Gateway: 172.17.0.254

DNS Server: 0.0.0.0

Start IP Address: 172 17 30 100

Subnet Mask: 255 255 0 0

Maximum Number of Users: 10

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	172.17.0.0	255.255....	0	0.0.0.0	0.0.0.0
DHCP_invalide	172.17.0...	0.0.0.0	172.17.3...	255.255....	10	0.0.0.0	0.0.0.0

☐ Top

PC 1

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address: 172.17.30.101

Subnet Mask: 255.255.0.0

Default Gateway: 172.17.0.254

DNS Server: 0.0.0.0

PC 2

IP Configuration	
☒ DHCP	☐ Static
DHCP request successful.	
IPv4 Address	172.17.30.100
Subnet Mask	255.255.0.0
Default Gateway	172.17.0.254
DNS Server	0.0.0.0

configurer le switch avec snooping

Activer le DHCP Snooping globalement sur le switch

```
ip dhcp snooping
```

Spécifier pour quel VLAN le snooping doit être actif

```
ip dhcp snooping vlan 1
```

Aller sur l'interface où est connecté le SERVEUR DHCP VALIDE

```
interface FastEthernet0/24
```

Déclarer ce port comme un port de confiance. Seuls les ports de confiance peuvent envoyer des messages DHCP

```
ip dhcp snooping trust
```

Limiter le nombre de paquets DHCP par seconde pour éviter les attaques par déni de service

```
ip dhcp snooping limit rate 100
```

Test avec les 2 dhcp

Avec les deux dhcp activée seul le dhcp valide est utilisé

PC1

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.100

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.254

DNS Server 0.0.0.0

PC2

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.101

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.254

DNS Server 0.0.0.0

Test avec le serveur DHCP valide éteint

le service DHCP n'est plus disponible

PC1

IP Configuration

☒ DHCP ☐ Static DHCP failed. APIPA is being used.

IPv4 Address 169.254.156.145

Subnet Mask 255.255.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

PC2

IP Configuration

☒ DHCP ☐ Static DHCP failed. APIPA is being used.

IPv4 Address 169.254.121.196

Subnet Mask 255.255.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

Vérification sur le switch

```
show ip dhcp snooping
```

on retrouve bien l'interface ou le snooping est actif et la limite de temps

```
Switch#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
1
Insertion of option 82 is enabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                Trusted      Rate limit (pps)
-----
FastEthernet0/3          no          unlimited
FastEthernet0/1          no          unlimited
FastEthernet0/2          no          unlimited
FastEthernet0/24         yes         100
Switch#
```

```
show ip dhcp snooping binding
```

on retrouve les 2 ip distribué

```
Switch#show ip dhcp snooping binding
MacAddress      IpAddress      Lease(sec)  Type           VLAN  Interface
-----
00:0D:BD:05:79:C4  192.168.1.101  86400      dhcp-snooping  1     FastEthernet0/1
00:09:7C:51:9C:91  192.168.1.102  86400      dhcp-snooping  1     FastEthernet0/2
Total number of bindings: 2
Switch#
```